How-To Library

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Importing Metadata from Cassandra and MongoDB Sources in Test Data Management

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Abstract

Import metadata from Cassandra and MongoDB data sources into Test Data Manager and perform Test Data Management (TDM) operations. This article describes how to configure data source names for Cassandra and MongoDB sources, import metadata into PowerCenter®, import metadata into TDM, and create ODBC connections for Cassandra and MongoDB sources.

Supported Versions

- Test Data Management 9.7.1 HotFix 1
- Test Data Management 10.1.0
- Test Data Management 10.2.0

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Overview

You can perform TDM operations on data from NoSQL databases such as Cassandra and MongoDB.

To perform TDM operations on source data, you must import the source metadata into the TDM repository. You cannot import metadata from a Cassandra or a MongoDB data source directly into Test Data Manager. You must import metadata from the PowerCenter repository.

Create connections to perform TDM operations on source data. Use an ODBC connection to create a Cassandra or a MongoDB connection in Test Data Manager.

When you create a project, add one or more sources to the project. You can perform data domain discovery and data masking operations on Cassandra sources. You can perform data discovery, data subset, data masking, and data generation operations on MongoDB sources. You can perform data subset operations on versions 10.1.0 with EBF-8803 and later versions. You can perform data generation operations on version 10.2.0. You can create data sets in the test data warehouse from data on MongoDB sources on TDM version 10.2.0.

Scenario

You work with a software product testing team that tests applications with test data.

You need to create test data from large volumes of production data on NoSQL databases such as Cassandra and MongoDB.

To create test data, you import the source metadata into Test Data Manager. Run a plan to create the test data.

To import metadata from a Cassandra or a MongoDB data source, perform the following tasks:

1. Configure a data source name (DSN) with Cassandra and MongoDB ODBC drivers on the PowerCenter Client machine.
2. Create a folder in the PowerCenter repository.
3. Import the source metadata into PowerCenter.
4. Create a project in Test Data Manager.
5. Import metadata from Cassandra and MongoDB sources.
6. Create an ODBC connection to connect to a Cassandra or a MongoDB data source in Test Data Manager. Use the connections in a plan to perform TDM operations.

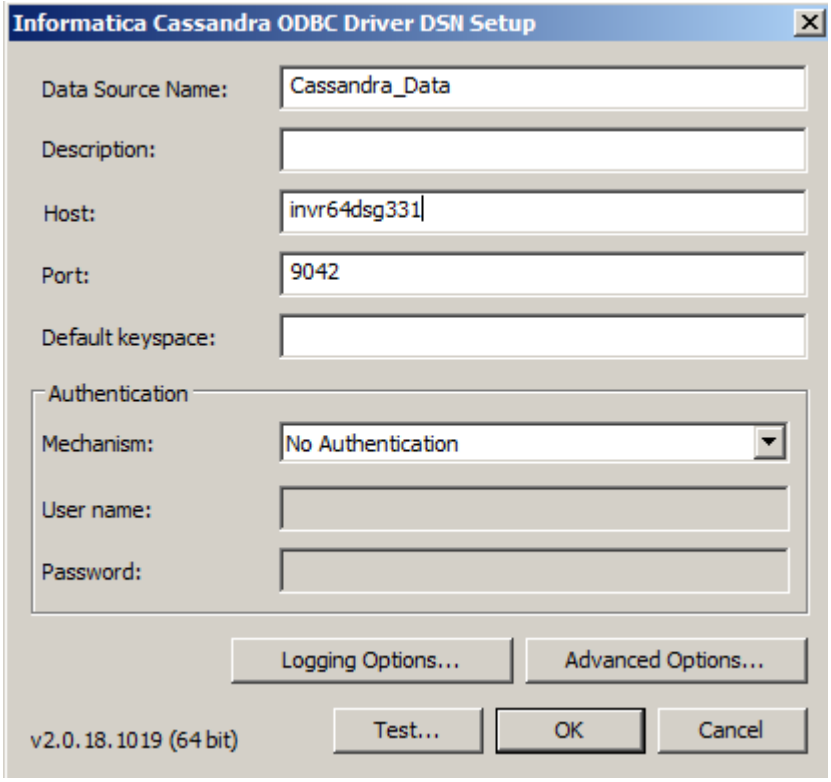
Configure Cassandra and MongoDB DSN

The PowerCenter Client requires an ODBC DSN to import metadata from Cassandra and MongoDB data sources. To import metadata into PowerCenter, configure an ODBC data source that connects to Cassandra and MongoDB drivers on the PowerCenter Client machine.

1. On the PowerCenter Client machine, open ODBC Data Source Administrator.
2. Click **System DSN** > **Add**.

   The **Create New Data Source** dialog box appears.
3. Select **Informatica Cassandra ODBC Driver** and click **Finish**.
4. In the **Informatica Cassandra ODBC Driver DSN Setup** dialog box, enter a data source name. Enter the host and port details.

   The following image shows the DSN properties that you can configure for Cassandra data sources:



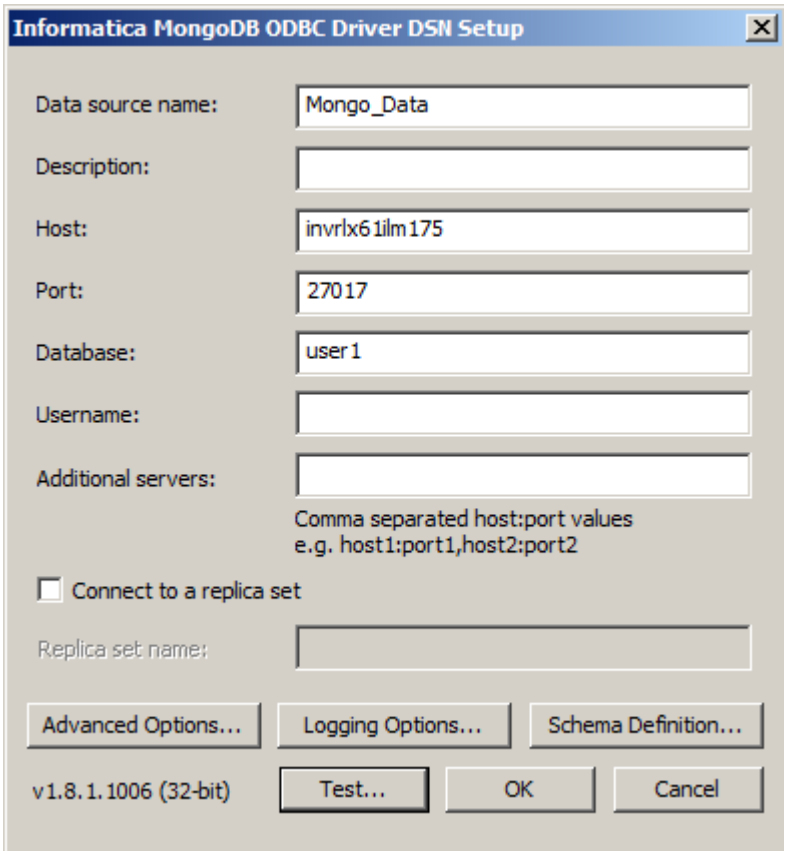
5. To validate the details, test the connection.

The Cassandra data source is created and appears in the list of system data sources.

6. Click **OK**.
7. To add the MongoDB DSN, click **Add**.

   The **Create New Data Source** dialog box appears.
8. Select **Informatica MongoDB ODBC Driver** and click **Finish**.
9. In the **Informatica MongoDB ODBC Driver DSN Setup** dialog box, enter a data source name. Enter the host, port, and database details.

   The following image shows the DSN properties that you can configure for MongoDB data sources:



10. To validate the details, test the connection.

    The MongoDB data source is created and appears in the list of system data sources.
11. Click **OK**.

Create a DSN Entry for Cassandra and MongoDB

The PowerCenter Integration Service requires a 64-bit data source to perform data movement operations. Configure 64-bit Cassandra and MongoDB ODBC data sources to access the databases to perform TDM operations. The task describes the steps to configure the data source name on a Linux machine.

Install the Cassandra and MongoDB 64-bit clients on the Linux machine on which you create the PowerCenter Integration Service.

1. Open the `<Informatica installation directory>\ODBC7.1\odbc.ini` file.

2. Optional. Add an entry for each data source that you create in the [ODBC Data Sources] section. For example, add the following in the [ODBC Data Sources] section:

```
Cassandra=libinformaticacassandraodbc64.so
MongoDB=libinformaticamongodbodbc64.so
```

3. Configure the data source properties to connect to Cassandra and MongoDB databases. You use the connections to perform TDM operations. Add an entry with the following properties for each Cassandra and MongoDB connection that you create:

```
[Cassandra_Data]
Description=Informatica CassandraDB ODBC Driver (64-bit) DSN
Driver=<Informatica installation directory>/tools/cassandra/lib/
libinformaticacassandraodbc64.so
Host=<Database server host>
Port=<Database port>
Database=<Database name>

[Mongo_Data]
Description=Informatica MongoDB ODBC Driver (64-bit) DSN
Driver=<Informatica installation directory>/tools/mongodb/lib/
libinformaticamongodbodbc64.so
Host=<Database server host>
Port=<Database port>
Database=<Database name>
```

4. Save and close the file.

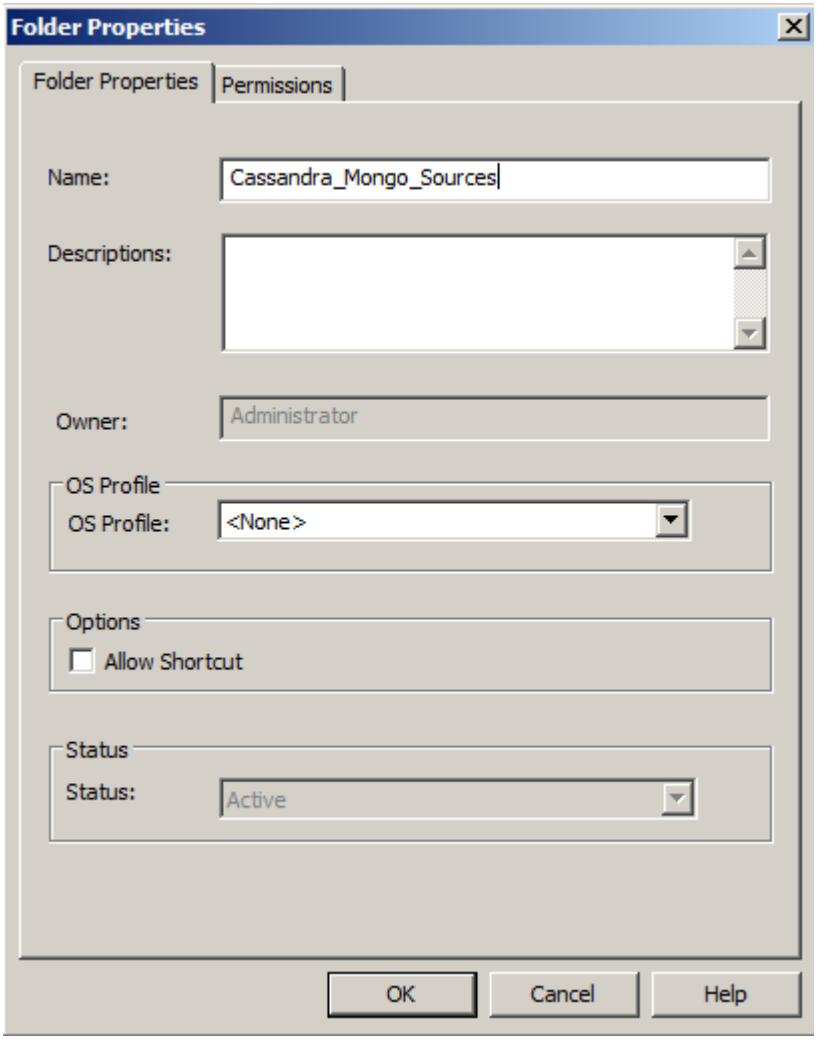
Create a Folder in the PowerCenter Repository

Create a folder in the PowerCenter repository where you can define the data sources and targets, build mappings, and run workflows. In this folder, you can import metadata from Cassandra and MongoDB data sources that you want to use for TDM operations.

1. In the Repository Manager, click **Folder** > **Create**.
2. Enter a name for the folder.

   By default, the user account that logged in is the owner of the folder and has full permissions on the folder.

The following image shows the folder properties in the Repository Manager:



3. Click **OK**.

   The Repository Manager displays a message that the folder has been successfully created.
4. Click **OK**.

   The folder appears in the repository.
5. Exit the Repository Manager.

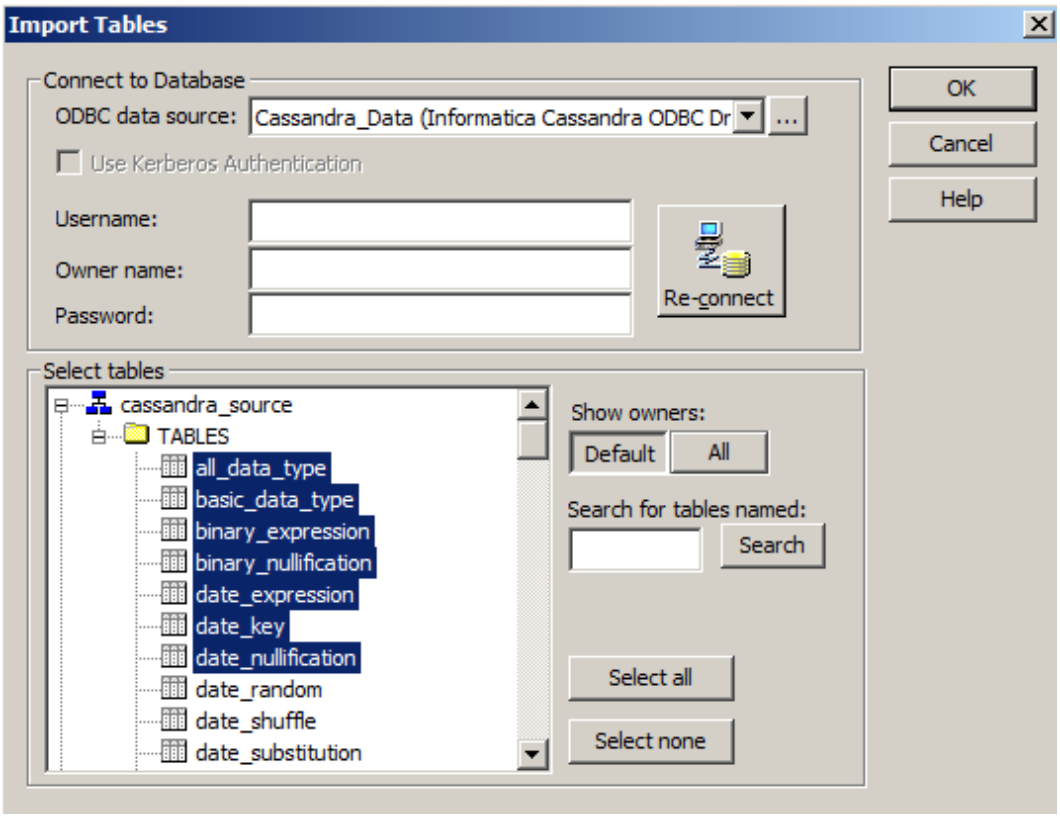
Import Cassandra and MongoDB Sources into PowerCenter

Import source metadata into PowerCenter from the Designer.

1. Open the Designer and connect to the repository.
2. Select the folder that you created.
3. To import metadata from the Cassandra data source, click **Sources** > **Import from Database**.

   The **Import Tables** dialog box appears.
4. Select the Cassandra ODBC data source that you created on the PowerCenter Client machine.

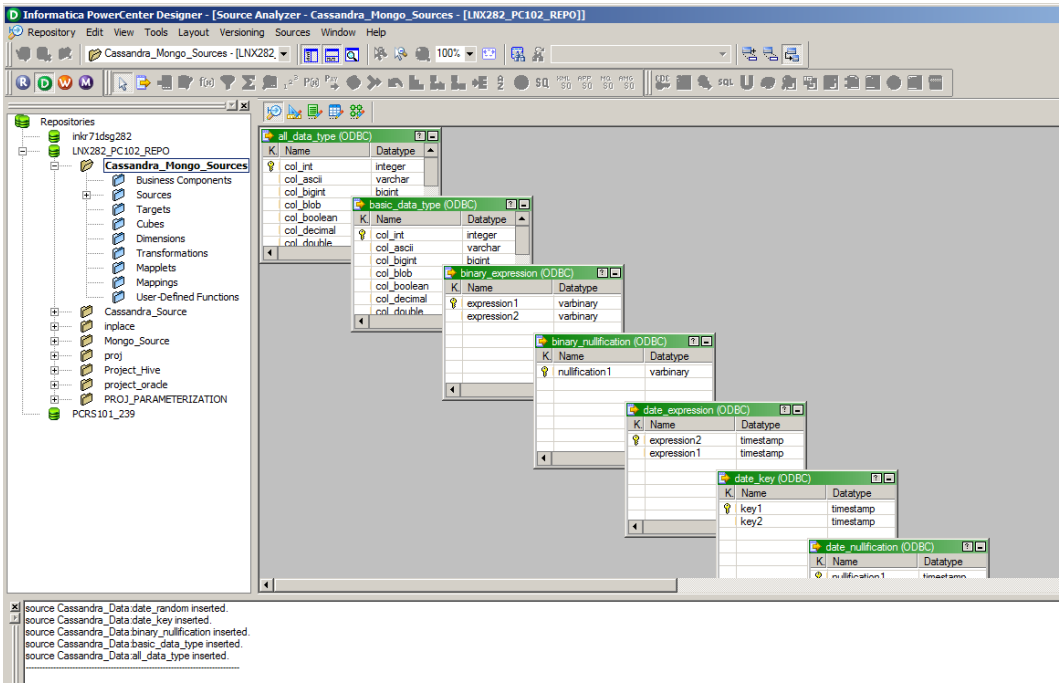
The following image shows a sample of Cassandra data source tables:



5. Click **Connect**. You do not need to enter the user name and password details.
6. Select the tables that you want to import and click **OK**.

   The tables are imported into the PowerCenter folder and open in the **Source Analyzer** window.

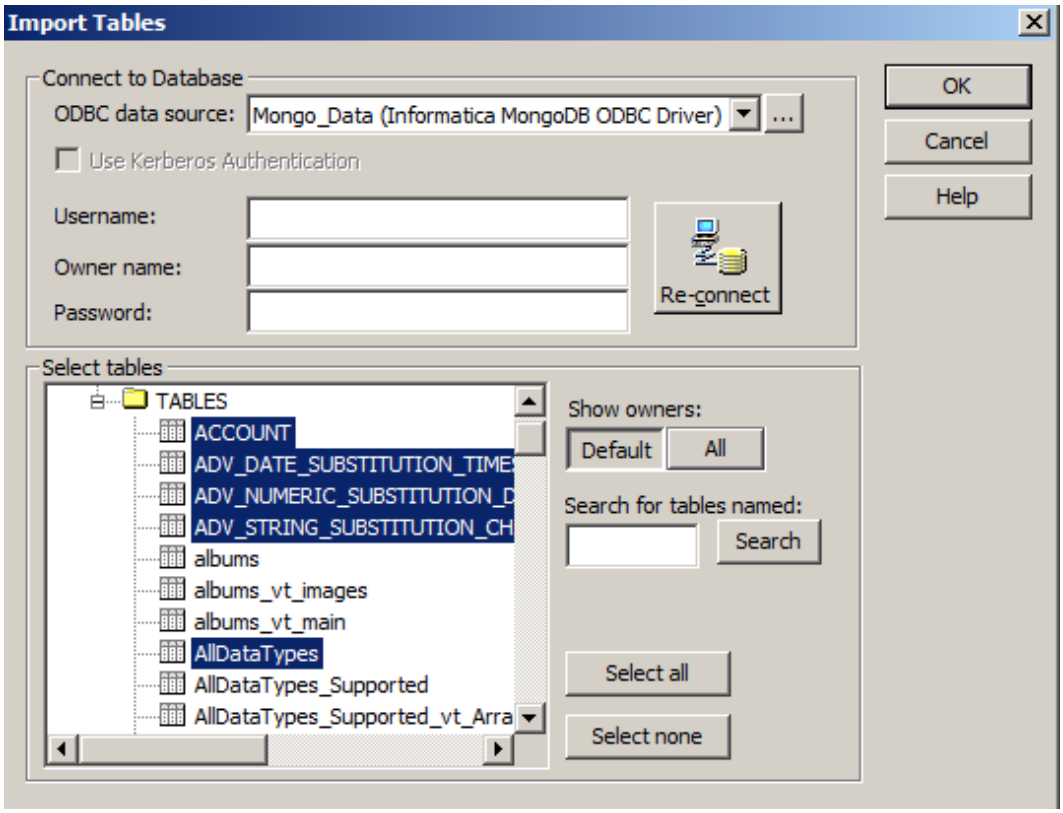
The following image shows the tables that you import from the Cassandra data source:



7. Save the source metadata that you imported.
8. To import metadata from the MongoDB data source, click **Sources** > **Import from Database**.

   The **Import Tables** window appears.
9. Select the MongoDB data source that you created on the PowerCenter Client machine.

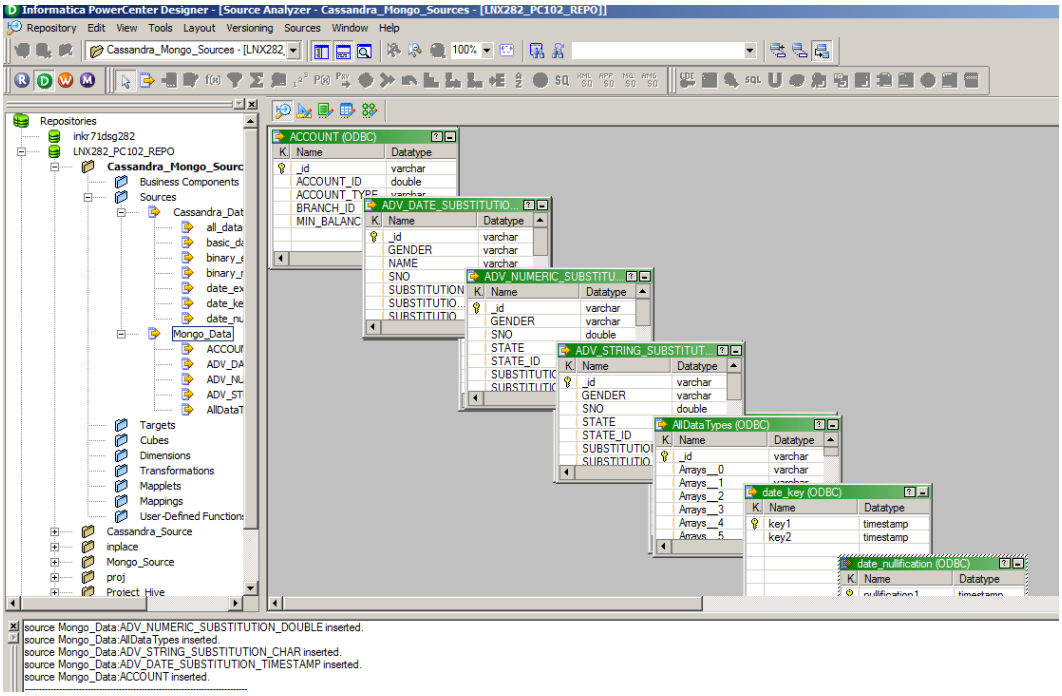
The following image shows a sample of MongoDB data source tables:



10. Click **Connect**. You do not need to enter the user name and password details.
11. Select the tables that you want to import and click **OK**.

    The tables are imported into the PowerCenter folder and open in the **Source Analyzer** window.

The following image shows the tables that you imported from the MongoDB data source:



12. Save the source metadata that you imported.

Create a Project in Test Data Manager

In Test Data Manager, create a project to store the TDM components.

1. In Test Data Manager, click **Projects**.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter the following project properties:

Name

Enter a name for the project.

Description

Optional. Enter a description.

PowerCenter Repository

TDM populates the name of the PowerCenter repository to store the repository folder.

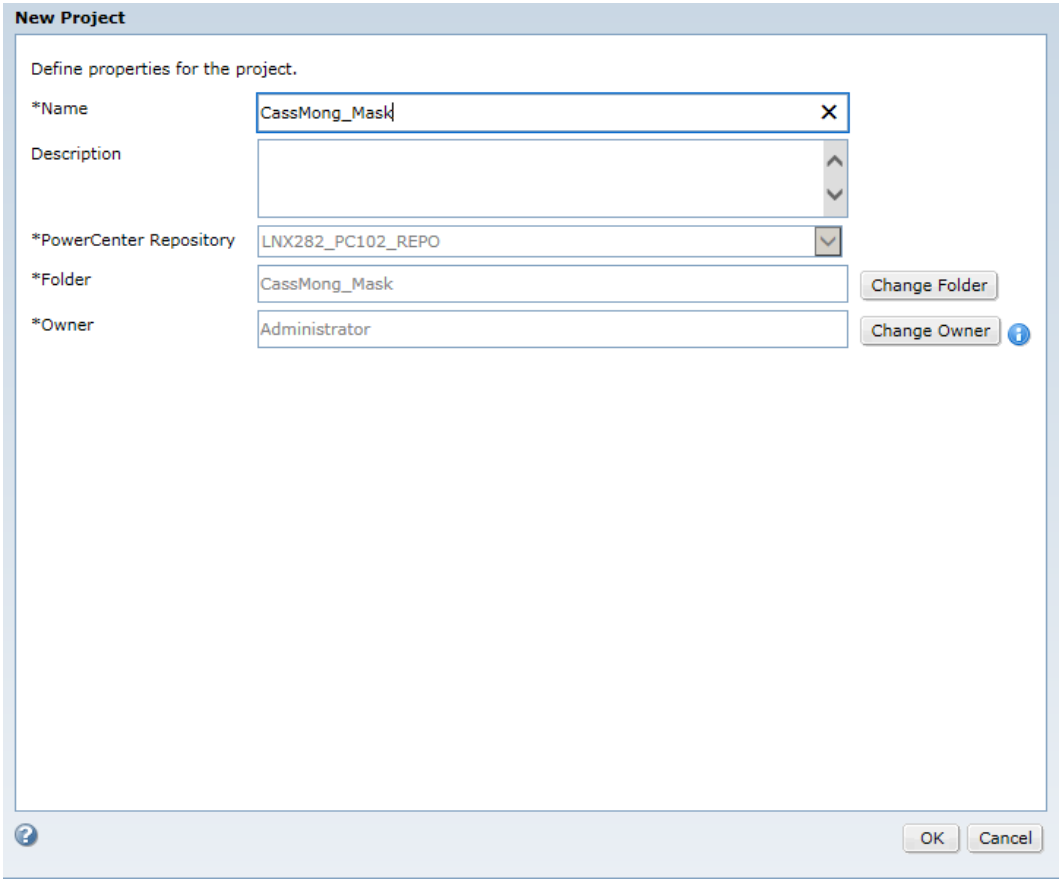
Folder

Default is the project name. You can choose another folder in the repository.

Owner

Default is the name of the user that created the folder. You can select another user as the folder owner.

The following image shows the project properties that you can enter:



4. Click **OK**.

   The project opens in a separate tab.

Import Metadata from Cassandra and MongoDB Sources

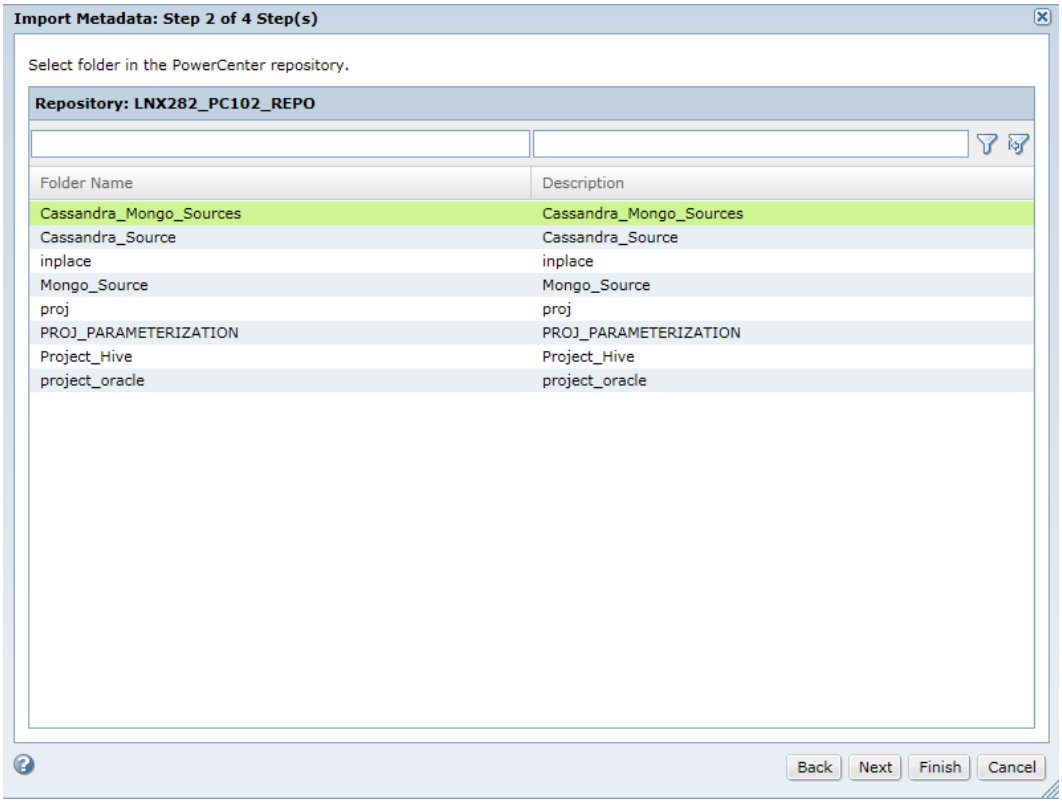
You cannot directly import metadata from a Cassandra or a MongoDB source in Test Data Manager. Import metadata from the PowerCenter repository to the project to perform TDM operations on the Cassandra and MongoDB data sources.

1. In the project, click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.

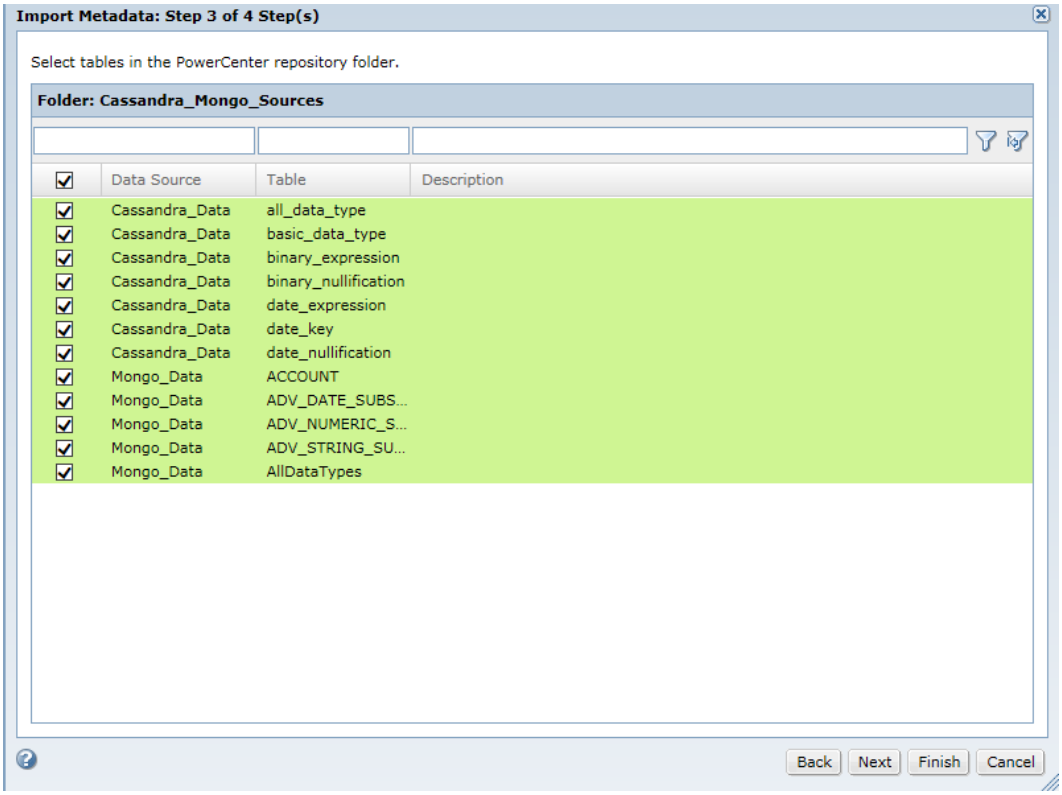
2. Choose to import metadata from the PowerCenter repository and click **Next**.
3. Select the folder into which you imported Cassandra and MongoDB metadata in the Designer and click **Next**.

The following image shows the list of folders in the PowerCenter repository:



4. Select the required tables.

The following image shows the list of tables present in Cassandra and MongoDB data sources:



5. Click **Next**.
6. To import the metadata immediately, choose **Import Now**.
7. Click **Finish**.

You can use the metadata from Cassandra and MongoDB data sources to perform TDM operations in Test Data Manager.

Create Cassandra and MongoDB Database Connections

To use Cassandra and MongoDB connections in a plan, create ODBC connections in Test Data Manager .

1. To create a connection, click **Administrator** > **Connections**
2. Click **Actions** > **New**.

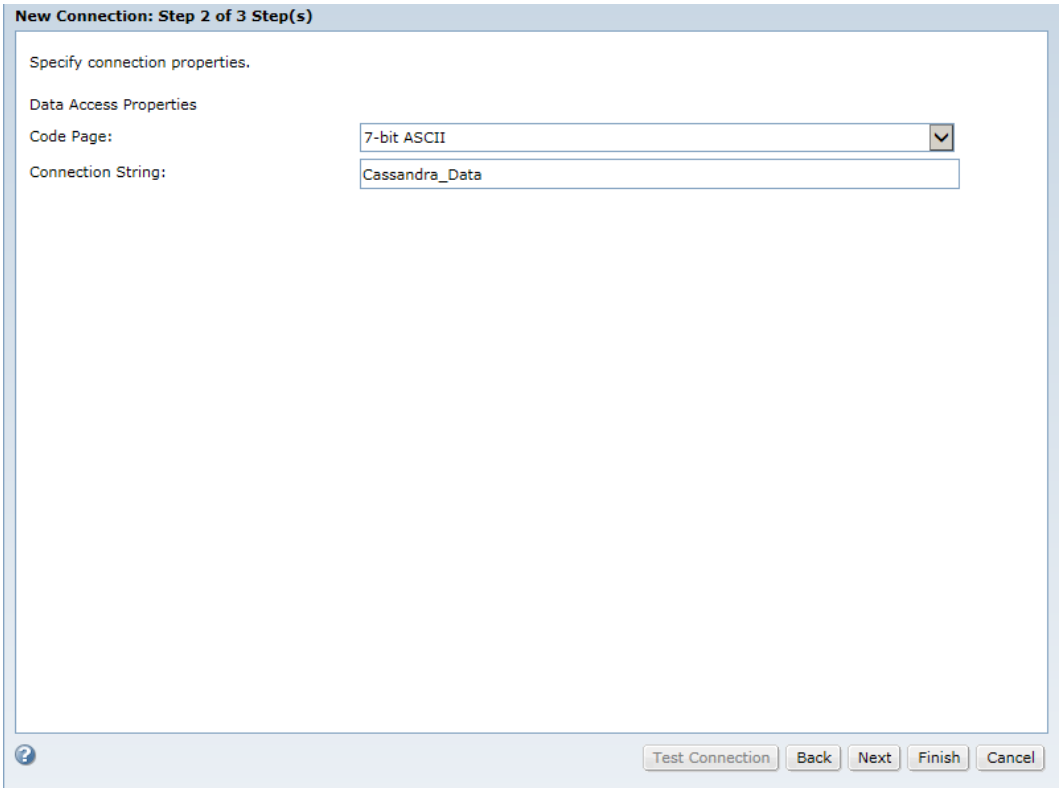
   A New Connection wizard appears.
3. To create a Cassandra data source connection, enter a connection name.

   The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that includes the connection might fail.
4. Select the ODBC connection type.
5. Click **Next**.

   The properties that appear depend on the type of ODBC connection that you select. For information about connection properties, see the *Test Data Management Administrator Guide*.
6. Enter the connection string for the Cassandra data source.

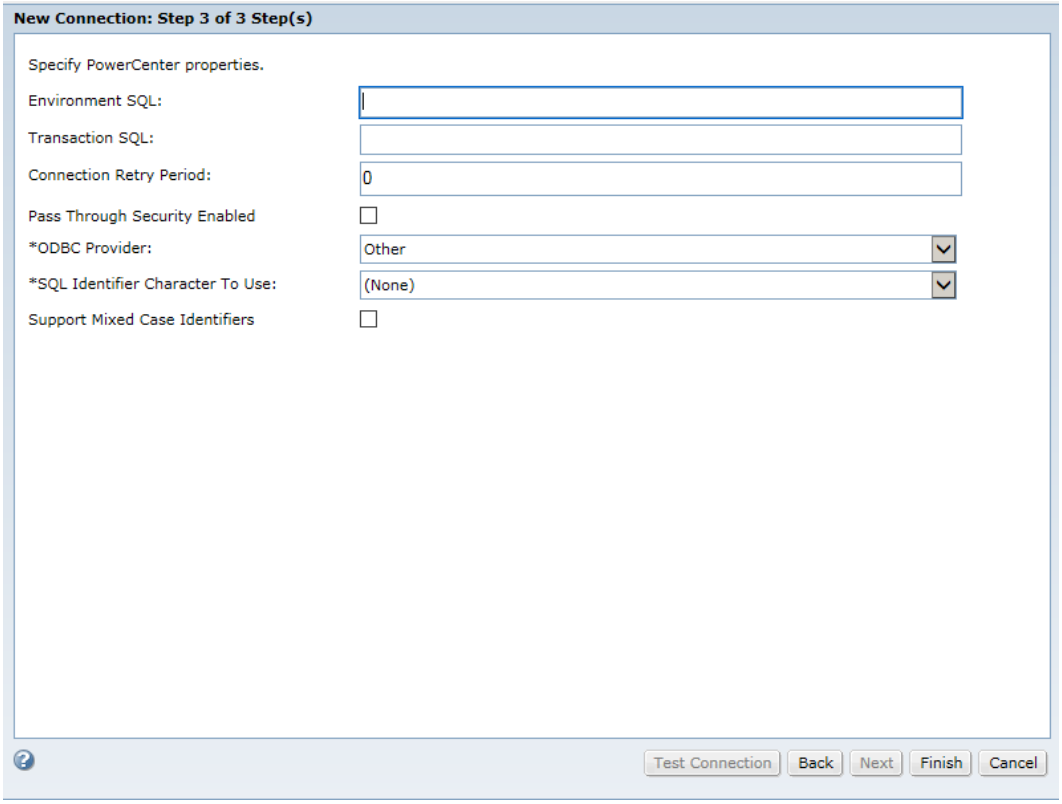
The connection string must be the name of the connection that you configured in the `odbc.ini` file.

The following image shows the data access properties for a connection:



7. Click **Next**.
8. In the ODBC Provider field, select **Other**.

The following image shows the PowerCenter properties that you can configure for the Cassandra connection:



9. To save the connection, click **Finish**.

   The connection is visible in the **Administrator** | **Connections** view.
   Follow the same steps to create a MongoDB connection.

You can use the connections in a plan.

Author

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